EXACTNESS PROPERTIES OF THE LOWER SHRIEK ALONG A MORPHISM OF SCHEMES ON BIG SITES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the listed properties; the section description of the lower shriek is used throughout and is recalled inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Lower shriek f_! for a morphism of big sites. Same chapter, the functor f_! on sheaves on the big site, right adjoint to the pullback f^{-1}, and its description by sections. Supplies the functor whose exactness properties are asserted.
- Comparison of the big site of X with the big site of its base and the morphism of topoi f_big. Same chapter, the morphism of topoi f_big and the description of the pullback and pushforward along it, together with the exactness of the pullback. Supplies the identifications with which the exactness statements are proved.

Source reference. Original lemma 0210: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7008–7008. Navigation pointer invoked at proof line 7008; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 021H: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7009–7009. Navigation pointer invoked at proof line 7009; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 04HC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7010–7010. Navigation pointer invoked at proof line 7010; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04HD: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7011–7011. Navigation pointer invoked at proof line 7011; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 021W: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7012–7012. Navigation pointer invoked at proof line 7012; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04BG: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7045–7045. Navigation pointer invoked at proof line 7045; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0797: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7047–7047. Navigation pointer invoked at proof line 7047; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original remark 0798: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7059–7059. Navigation pointer invoked at proof line 7059; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04BH: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7061–7061. Navigation pointer invoked at proof line 7061; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\tau \in \{Zariski, étale, smooth, syntomic, fppf\}$. Let $f : X \to Y$ be a morphism of schemes. Let*

$$f_{big} : Sh((Sch/X)_\tau) \longrightarrow Sh((Sch/Y)_\tau)$$

be the corresponding morphism of topoi as in Topologies, Lemma 0210, 021H, 04HC, 04HD, or 021W.

(1) *The functor $f_{big}^{-1} : Ab((Sch/Y)_\tau) \to Ab((Sch/X)_\tau)$ has a left adjoint*

$$f_{big!} : Ab((Sch/X)_\tau) \to Ab((Sch/Y)_\tau)$$

which is exact.

(2) *The functor $f_{big}^* : Mod((Sch/Y)_\tau, \mathcal{O}) \to Mod((Sch/X)_\tau, \mathcal{O})$ has a left adjoint*

$$f_{big!} : Mod((Sch/X)_\tau, \mathcal{O}) \to Mod((Sch/Y)_\tau, \mathcal{O})$$

which is exact.

Moreover, the two functors $f_{big!}$ agree on underlying sheaves of abelian groups.

Proof. Recall that f_{big} is the morphism of topoi associated to the continuous and cocontinuous functor $u : (Sch/X)_\tau \to (Sch/Y)_\tau$, $U/X \mapsto U/Y$. Moreover, we have $f_{big}^{-1}\mathcal{O} = \mathcal{O}$. Hence the existence of $f_{big!}$ follows from Modules on Sites, Lemma 04BG, respectively Modules on Sites, Lemma 0797. Note that if U is an object of $(Sch/X)_\tau$ then the functor u induces an equivalence of categories

$$u' : (Sch/X)_\tau/U \longrightarrow (Sch/Y)_\tau/U$$

because both sides of the arrow are equal to $(Sch/U)_\tau$. Hence the agreement of $f_{big!}$ on underlying abelian sheaves follows from the discussion in Modules on Sites, Remark 0798. The exactness of $f_{big!}$ follows from Modules on Sites, Lemma 04BH as the functor u above which commutes with fibre products and equalizers. □